WORKED SOLUTION

STEP III 2019, Question 11

Probability and Statistics • Poisson Thinning

STEP III 2019, Question 11

Read the question before you go on. It is in the STEP III paper in OCR's download of the 2019 STEP papers (a zip file):

www.ocr.org.uk/Images/696590-2019-past-papers.zip

The key idea

Everything rests on conditioning on the number of customers who arrive, or who take sand, and then summing a Poisson series. Part (i) is the classic "thinning" of a Poisson distribution: when the binomial and Poisson probabilities are multiplied together, the sum over the unseen total collapses to an exponential. Once we know that the number of sand-takers is Poisson, parts (ii) and (iii) only need the expectation of $(1-k)^Y$, and that is another exponential series.

Solution

Let N be the number of customers in a day, so $N \sim \mathrm{Po}(\lambda)$, and let Y be the number who take sand. We assume customers decide independently of each other and of N, so that given $N = n$ we have $Y \sim \mathrm{B}(n, p)$.

Part (i)

For $y = 0, 1, 2, \ldots$, conditioning on N (only $n \geq y$ contributes):

$$\begin{aligned}
\mathrm{P}(Y = y) &= \sum_{n=y}^{\infty} \mathrm{e}^{-\lambda} \frac{\lambda^n}{n!} \binom{n}{y} p^y (1-p)^{n-y} \\
&= \mathrm{e}^{-\lambda} \frac{(p\lambda)^y}{y!} \sum_{n=y}^{\infty} \frac{(\lambda(1-p))^{n-y}}{(n-y)!} \qquad \text{since } \frac{\lambda^n}{n!} \cdot \frac{n!}{y!\,(n-y)!} = \frac{\lambda^y \lambda^{n-y}}{y!\,(n-y)!} \\
&= \mathrm{e}^{-\lambda} \frac{(p\lambda)^y}{y!} \mathrm{e}^{\lambda(1-p)} = \mathrm{e}^{-p\lambda} \frac{(p\lambda)^y}{y!},
\end{aligned}$$

where we put $m = n - y$ and used $\sum_{m \geq 0} x^m / m! = \mathrm{e}^x$. This is the $\mathrm{Po}(p\lambda)$ probability, so $Y \sim \mathrm{Po}(p\lambda)$, as required.

Part (ii)

Each customer who takes sand leaves a fraction $1-k$ of what was there. So after Y sand-takers the mass left is $S(1-k)^Y$, and the mass taken is

$$T = S(1-(1-k)^Y).$$

By part (i),

$$\mathrm{E}[(1-k)^Y] = \sum_{y=0}^{\infty} (1-k)^y \, \mathrm{e}^{-p\lambda} \frac{(p\lambda)^y}{y!} = \mathrm{e}^{-p\lambda} \, \mathrm{e}^{p\lambda(1-k)} = \mathrm{e}^{-kp\lambda}.$$

Hence, by linearity of expectation,

$$\mathrm{E}(T) = S(1-\mathrm{E}[(1-k)^Y]) = (1-\mathrm{e}^{-kp\lambda})S,$$

which is the required result.

Part (iii)

> **Ask yourself** How do we turn "a proportion k of the sand" into a probability about one particular grain?

Since there are a great many grains and none is singled out, whenever someone takes a proportion k of the remaining sand, a grain that is still in the bag is taken with probability k, independently of what happened earlier. (Equivalently, the chance that the golden grain is in a given portion equals that portion's share of the mass.)

Let G be the event that the assistant takes the golden grain. Given $Y = y$, the grain survives all y customers with probability $(1-k)^y$, and then the assistant takes it with probability k. So

$$\mathrm{P}(G \mid Y = y) = k(1-k)^y,$$

and, using the expectation found in part (ii),

$$\mathrm{P}(G) = k\,\mathrm{E}[(1-k)^Y] = k\,\mathrm{e}^{-kp\lambda}.$$

This agrees with part (ii): the probability that the golden grain survives the day is $\mathrm{e}^{-kp\lambda}$, which is exactly the expected fraction of sand remaining.

The case $k = 0$. Then $\mathrm{P}(G) = 0$. This makes sense: with $k = 0$ nobody, the assistant included, takes any sand, so the grain stays in the bag.

The limit $k \to 1$. Then $\mathrm{P}(G) \to \mathrm{e}^{-p\lambda}$, which is $\mathrm{P}(Y = 0)$. When k is close to 1, the first customer who takes sand takes nearly all of it, golden grain almost certainly included. The assistant essentially gets the grain only if no customer took sand, and then takes (almost) the whole bag.

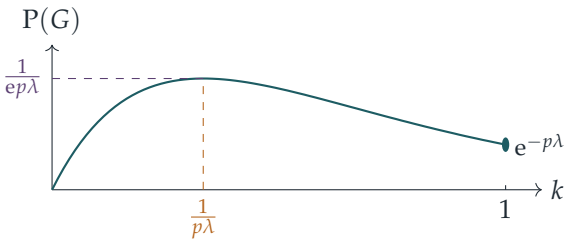
Maximising. Write $\mu = p\lambda$ and $f(k) = k\,\mathrm{e}^{-\mu k}$ for $0 \le k < 1$. Then

$$f'(k) = \mathrm{e}^{-\mu k}(1-\mu k),$$

which is positive for $k < 1/\mu$ and negative for $k > 1/\mu$. Since $\mu > 1$, the point $k = 1/\mu$ lies in $(0,1)$, so f increases up to it and decreases after it, and it gives the maximum over $[0,1)$. Hence the probability is maximised when

$$k = \frac{1}{p\lambda},$$

and the maximum probability is $\dfrac{1}{\mathrm{e}\,p\lambda}$. (If $p\lambda \le 1$ then f increases throughout $[0,1)$ and there is no maximum, only the supremum $\mathrm{e}^{-p\lambda}$ as $k \to 1$. That is why the question restricts to $p\lambda > 1$.)



The graph shows $\mathrm{P}(G) = k\,\mathrm{e}^{-kp\lambda}$ in the case $p\lambda = 3$.

Checks

- Part (i) passes the obvious tests: $p = 1$ gives back $\mathrm{Po}(\lambda)$, and $p = 0$ gives $Y = 0$ for certain.
- In part (ii), $k = 0$ gives zero sand taken, and as $\lambda \to \infty$ the expected mass taken tends to S, as it should.
- With $\lambda = 3$, $p = 0.6$, $k = 0.4$ the formula gives $\mathrm{P}(G) = 0.4\,\mathrm{e}^{-0.72} \approx 0.1947$. A computer simulation of 200 000 days gave 0.1953.
- With $p\lambda = 3$, the largest value of $k\,\mathrm{e}^{-3k}$ on a fine grid occurs at $k \approx 0.3333$ and equals $0.1226 \approx 1/(3\mathrm{e})$.

Common slips

- Writing the sand taken as kSY, as though each customer took a proportion k of the *original* sand. Each customer takes a share of what is *left*, so the amounts form a geometric sequence.
- Computing $\mathrm{E}[(1-k)^Y]$ as $(1-k)^{\mathrm{E}(Y)} = (1-k)^{p\lambda}$. The expectation of a non-linear function is not the function of the expectation.
- Forgetting to check that $1/(p\lambda)$ actually lies in the allowed range $0 \le k < 1$. This is exactly where the condition $p\lambda > 1$ is used.

Beyond STEP

Part (i) is the thinning property of the Poisson distribution. The expectation $\mathrm{E}[s^Y] = \mathrm{e}^{\mu(s-1)}$ that appears in parts (ii) and (iii) is the *probability generating function* of $\mathrm{Po}(\mu)$, evaluated at $s = 1 - k$. Generating functions make thinning almost a one-liner: if $\mathrm{E}[s^N] = \mathrm{e}^{\lambda(s-1)}$, then $\mathrm{E}[s^Y] = \mathrm{E}[(1-p+ps)^N] = \mathrm{e}^{\lambda p(s-1)}$. In the theory of Poisson processes this becomes the statement that independently marking the points of a Poisson process gives Poisson processes again. Grimmett and Welsh, *Probability: An Introduction*, treats generating functions and these ideas clearly.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.